Tecnológico de Estudios Superiores de Villa Guerrero

Ingeniería en Sistemas Computacionales

Ingenieria del software

Reporte 2

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Grupo:
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Inspección de la Interfaz Local (Get-NetIPAddress) Antes de salir a la red, debemos conocer nuestra identidad en ella. Get-NetIPAddress -AddressFamily IPv4 | Select-Object InterfaceAlias, IPAddress, PrefixLength

```
C:\Program Files\WindowsAp

PowerShell 7.6.1
PS C:\Windows\System32> Get-NetIPAddress -AddressFamily IPv4 | Select-Object InterfaceAlias,
>> IPAddress, PrefixLength

InterfaceAlias              IPAddress       PrefixLength
--------------              ---------       ------------
Ethernet                    192.168.56.1              24
Wi-Fi 2                     10.61.209.102             24
Conexión de área local* 2   169.254.54.28             16
Conexión de red Bluetooth   169.254.229.204           16
Conexión de área local* 1   169.254.56.222            16
Loopback Pseudo-Interface 1 127.0.0.1                  8

PS C:\Windows\System32>
```

Verificación de Conectividad (Test-Connection vs ping) Aunque ping es el estándar clásico, Test-Connection devuelve objetos que pueden ser procesados. # Uso clásico ping google.com # Uso avanzado en PowerShell (3 paquetes) Test-Connection -ComputerName google.com -Count 3

```
PS C:\Windows\System32> Test-Connection -ComputerName google.com -Count 3

   Destination: google.com

Ping Source           Address                  Latency BufferSize Status
                                                  (ms)        (B)
---- ------           -------                  ------- ---------- ------
   1 DESKTOP-NJ74HIN  142.250.26.113               125         32 Success
   2 DESKTOP-NJ74HIN  142.250.26.113               107         32 Success
   3 DESKTOP-NJ74HIN  142.250.26.113               233         32 Success

PS C:\Windows\System32>
```

Diagnóstico de Ruta y Puerto (Test-NetConnection) Este es uno de los comandos más potentes para un administrador. • Ejercicio (Trazado de ruta): Test-NetConnection -ComputerName 8.8.8.8 -TraceRoute

```
PS C:\Windows\System32> Test-NetConnection -ComputerName 8.8.8.8 -TraceRoute
TraceRoute [ICMP Echo Request (Max TTL = 30)]
```

Resolución de Nombres (Resolve-DnsName) Resolve-DnsName -Name tecvillaguerrero.edu.mx -Type A

```
PS C:\Windows\System32> Test-NetConnection -ComputerName google.com -Port 443

ComputerName     : google.com
RemoteAddress    : 142.250.26.113
RemotePort       : 443
InterfaceAlias   : Wi-Fi 2
SourceAddress    : 10.61.209.102
TcpTestSucceeded : True

PS C:\Windows\System32>
```

Análisis del Enrutamiento (Get-NetRoute) La tabla de enrutamiento decide por dónde sale cada paquete de datos. Get-NetRoute -AddressFamily IPv4

```
PS C:\Windows\System32> Resolve-DnsName -Name tecvillaguerrero.edu.mx -Type A
Resolve-DnsName: tecvillaguerrero.edu.mx : El nombre DNS no existe.
PS C:\Windows\System32>
```

Monitoreo de Sockets y Puertos (Get-NetTCPConnection) Get-NetTCPConnection -State Established | Select-Object LocalAddress, LocalPort, RemoteAddress, RemotePort, Stat

```
PS C:\Windows\System32> Get-NetRoute -AddressFamily IPv4

ifIndex DestinationPrefix                              NextHop                                   RouteMetric ifMetric PolicyStore
------- -----------------                              -------                                   ----------- -------- -----------
19      255.255.255.255/32                             0.0.0.0                                           256 25       ActiveStore
14      255.255.255.255/32                             0.0.0.0                                           256 25       ActiveStore
5       255.255.255.255/32                             0.0.0.0                                           256 55       ActiveStore
20      255.255.255.255/32                             0.0.0.0                                           256 65       ActiveStore
18      255.255.255.255/32                             0.0.0.0                                           256 25       ActiveStore
1       255.255.255.255/32                             0.0.0.0                                           256 75       ActiveStore
19      224.0.0.0/4                                    0.0.0.0                                           256 25       ActiveStore
14      224.0.0.0/4                                    0.0.0.0                                           256 25       ActiveStore
5       224.0.0.0/4                                    0.0.0.0                                           256 55       ActiveStore
20      224.0.0.0/4                                    0.0.0.0                                           256 65       ActiveStore
18      224.0.0.0/4                                    0.0.0.0                                           256 25       ActiveStore
1       224.0.0.0/4                                    0.0.0.0                                           256 75       ActiveStore
18      192.168.56.255/32                              0.0.0.0                                           256 25       ActiveStore
18      192.168.56.1/32                                0.0.0.0                                           256 25       ActiveStore
18      192.168.56.0/24                                0.0.0.0                                           256 25       ActiveStore
1       127.255.255.255/32                             0.0.0.0                                           256 75       ActiveStore
1       127.0.0.1/32                                   0.0.0.0                                           256 75       ActiveStore
1       127.0.0.0/8                                    0.0.0.0                                           256 75       ActiveStore
5       10.61.209.255/32                               0.0.0.0                                           256 55       ActiveStore
5       10.61.209.102/32                               0.0.0.0                                           256 55       ActiveStore
5       10.61.209.0/24                                 0.0.0.0                                           256 55       ActiveStore
1       10.3.0.20/32                                   0.0.0.0                                             1 75       ActiveStore
5       0.0.0.0/0                                      10.61.209.199                                       0 55       ActiveStore

PS C:\Windows\System32>
```

```
PS C:\Windows\System32> Get-NetTCPConnection -State Established | Select-Object LocalAddress,
>> LocalPort, RemoteAddress, RemotePort, State

LocalAddress  : 10.61.209.102
LocalPort     : 65026
RemoteAddress : 35.186.224.36
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 64646
RemoteAddress : 52.123.128.14
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 64416
RemoteAddress : 52.123.129.14
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 62752
RemoteAddress : 35.186.224.31
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 61651
RemoteAddress : 52.123.129.14
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
```

```
LocalAddress  : 10.61.209.102
LocalPort     : 60529
RemoteAddress : 172.217.72.188
RemotePort    : 5228
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 59898
RemoteAddress : 52.188.247.146
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 59897
RemoteAddress : 157.240.244.60
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 58907
RemoteAddress : 35.186.224.36
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 57813
RemoteAddress : 52.123.188.101
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 57606
RemoteAddress : 35.186.224.24
RemotePort    : 443
State         : Established
```

```
LocalAddress  : 10.61.209.102
LocalPort     : 50117
RemoteAddress : 157.240.14.52
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 49416
RemoteAddress : 104.208.203.89
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 49415
RemoteAddress : 104.208.203.89
RemotePort    : 443
State         : Established

LocalAddress  : 10.61.209.102
LocalPort     : 49251
RemoteAddress : 190.123.42.96
RemotePort    : 443
State         : Established

LocalAddress  : 127.0.0.1
LocalPort     : 7768
RemoteAddress : 127.0.0.1
RemotePort    : 55297
State         : Established

PS C:\Windows\System32>
```